Difference-in-Differences

Beyond the Switch: Doses, Treatments that Turn On and Off

F. Candau (UPPA-TREE)

Sum up, and what comes next

The two previous chapters had a treatment that is a **switch**: zero, then one, and one for good. One date (chapter 4), or one date per unit (chapter 5).

Two departures from the switch occur in much applied work:

1. the treatment is a **dose** — a tax rate, an aid amount per hectare, an exposure share;
2. the treatment **turns on and off** — a ban lifted by derogation then reinstated, a temporary programme, a threshold crossed in both directions.

This chapter

Each departure changes the potential outcomes, the parameter, and the parallel-trends assumption. The regression with two-way fixed effects has the same form in both cases, so it does not display the change. The chapter writes out what changes in each case and what the corresponding estimator computes.

Roadmap

1. **Doses —** potential outcomes indexed by a dose; the level effect $\mathrm{ATT}(d \mid d)$ versus the causal response; what strong parallel trends identify; what the two-way fixed-effects coefficient mixes.
2. **On and off —** treatment sequences; the curse of dimensionality; no-carryover; the switchers estimator of de Chaisemartin and D'Haultfœuille and its event-study version.
3. **Workflow —** the same forward engineering, twice.

Reference for the whole chapter: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (2026), Appendix A; Callaway, Goodman-Bacon & Sant'Anna (2024); de Chaisemartin & D'Haultfœuille (2020, 2024).

Section 1

Doses

Potential outcomes with a dose

Two periods; nobody treated in period 1; in period 2 unit i receives a dose $D_i \geq 0$ (possibly zero). The potential outcome $Y_{it}(d)$ is the outcome under dose d; $Y_{it}(0)$ is the untreated outcome. **Groups are defined by the dose.**

- Examples: a per-hectare aid amount; the share of a farm's land in a regulated zone; the exposure share of chapter 3.
- Level parameters:

  $$\mathrm{ATT}(d \mid d) = \mathbb{E}\left[Y_2(d) - Y_2(0) \mid D = d\right], \qquad \mathrm{ATE}(d) = \mathbb{E}\left[Y_2(d) - Y_2(0)\right].$$

  The first is the effect of dose d on the units that received d; the second, on a unit drawn at random.
- **Average causal response** (ACR): the effect of a marginal increment of the dose; ACRT, its version on the treated:

  $$\mathrm{ACR}(d) = \frac{\partial\, \mathrm{ATE}(d)}{\partial d}, \qquad \mathrm{ACRT}(d \mid d) = \frac{\partial\, \mathrm{ATT}(l \mid d)}{\partial l}\bigg|_{l=d},$$

  where $\mathrm{ATT}(l \mid d) = \mathbb{E}[Y_2(l) - Y_2(0) \mid D = d]$ is the effect of dose l on the units that received d (Angrist and Imbens 1995 for variable treatment intensity).

Two parallel-trends assumptions

Parallel trends, dose version

For every dose d: $\mathbb{E}[\Delta Y(0) \mid D = d] = \mathbb{E}[\Delta Y(0) \mid D = 0]$ — the units at dose d would have moved like the untreated units.

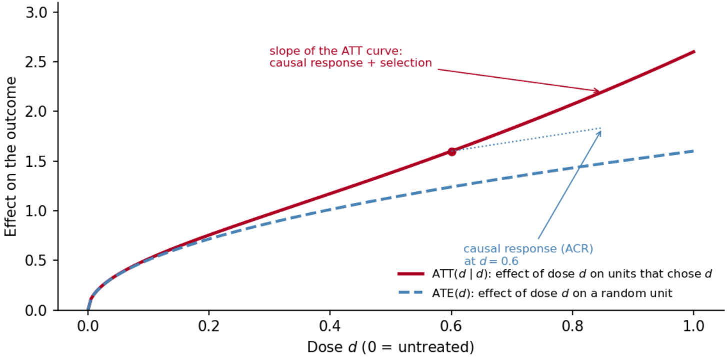
Under it, $\mathrm{ATT}(d \mid d)$ and the whole curve $d \mapsto \mathrm{ATT}(d \mid d)$ are identified as in the binary case (Callaway et al. 2024).

The slope of that curve contains selection

$\mathrm{ATT}(d' \mid d') - \mathrm{ATT}(d \mid d)$ compares two different groups of units; those who chose a higher dose may have a higher return to it: $\frac{d}{dd}\mathrm{ATT}(d \mid d) = \mathrm{ACRT}(d \mid d) + \partial \mathrm{ATT}(d \mid d')/\partial d' \big|_{d'=d}$, the second term being **selection into dose**.

$\mathrm{ATE}(d)$, $\mathrm{ACR}(d)$ and comparisons across doses require **strong parallel trends**, $\mathbb{E}[\Delta Y(d) \mid D = d'] = \mathbb{E}[\Delta Y(d) \mid D = d]$ for all d, d': every group would have responded to dose d alike, an assumption on *treated* potential outcomes. Then $\mathrm{ATT}(d \mid d) = \mathrm{ATE}(d)$ and the slope of the curve is the causal response.

The curve and its slope



Simulated: units with the steepest response choose the highest dose. Solid curve: $\mathrm{ATT}(d \mid d)$, identified under parallel trends; dashed: $\mathrm{ATE}(d)$; dotted segment: the causal response at $d = 0.6$. The slope of the solid curve adds the selection term to the causal response. Under strong parallel trends the two curves coincide.

What the two-way fixed-effects coefficient mixes

The natural regression puts the dose on the right-hand side:

$$y_{it} = f_i + f_t + \beta\, D_i \cdot \mathrm{Post}_t + u_{it}.$$

- With a binary treatment, β is the 2×2 DiD. With a dose, β is a **weighted average of causal responses across doses** plus selection terms (Callaway et al. 2024: several decompositions exist, “all having important limitations as meaningful summaries of treatment effects”).
- The weights come from the variance of the dose, as in chapter 5 they came from the variance of the timing: doses far from the mean weigh most.
- Under strong parallel trends the selection terms vanish and β is a dose-variance weighted average of $\mathrm{ACR}(d)$.

Forward engineering, dose version

Report the level curve $\mathrm{ATT}(d \mid d)$ (binned doses) and its aggregate $\mathbb{E}[\mathrm{ATT}(D \mid D) \mid D > 0]$; report the causal response separately, with the strong parallel-trends assumption written next to it.

Two further cases

- **Everybody already treated in period 1** at some level, doses change over time (a tax rate revised every year). de Chaisemartin et al. (2024), a different paper from the event study of the next section, identify per-dose effects for **switchers** when a sizable set of units keep their dose (**stayers**) and past doses do not affect current outcomes. Without stayers the comparison group is empty and DiD identifies nothing without further structure.
- **Functional form.** A dose-response in levels and one in logs impose different parallel trends; the two cannot both hold unless trends are trivial. State the scale of the outcome as part of the assumption (Roth and Sant'Anna 2023).
- **Diagnostics** carry over dose by dose: pre-trends of each dose bin against the untreated, and a Rambachan–Roth bound on the level curve.

Section 2

On and off

Treatment sequences

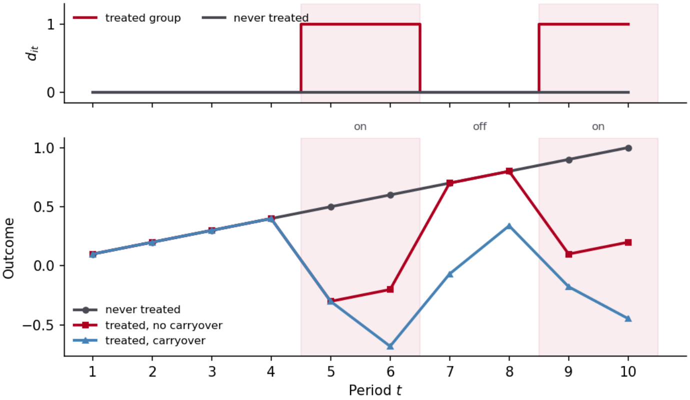
Treatment can be present in some periods and absent in others. The unit's treatment is a **path** $\mathbf{d} = (d_{i1}, \ldots, d_{iT}) \in \{0,1\}^T$, and the potential outcome $Y_{it}(\mathbf{d})$ depends on the whole path. **Groups are defined by paths.**

- Three periods, nobody treated in period 1: four paths, $(0,0,0)$, $(0,0,1)$, $(0,1,1)$ and $(0,1,0)$ — never, late, early, and **treated once then not**. Staggered adoption is the special case where the path is determined by its first 1.
- Building block: $\mathrm{ATT}(\mathbf{d}, t) = \mathbb{E}[Y_t(\mathbf{d}) - Y_t(\mathbf{0}) \mid G = \mathbf{d}]$, the effect at t of the path $\mathbf{d}$ against never being treated, for the units on that path (G = the unit's group, here its path); identified as in chapter 5 by comparison with never- or not-yet-treated units.

The curse of dimensionality

With T periods there are up to 2^{T-1} paths, each a group, most of them small. The literature restricts **carryover**, the effect of past treatment on current outcomes, and targets aggregated parameters.

Carryover: what the switch-off identifies



A ban in force in periods 5–6, lifted in 7–8, reinstated in 9–10. Without carryover the outcome returns to the untreated path as soon as the ban is lifted; with carryover the off period shows a lagged effect. The off period identifies the persistence of the effect, provided the estimand allows for it; the estimand of de Chaisemartin and D'Haultfœuille (2024), two slides below, does.

Three ways to shrink the problem

Paper	Restriction	What is estimated
de Chaisemartin & D'Haultfœuille (2020)	**no carryover**: the effect lasts one period	the instantaneous effect of the current treatment: units that *switch* at t against units whose treatment does not change (DID_M in their notation)
Imai, Kim & Wang (2023); Liu, Wang & Xu (2024)	**limited carryover**: at most k periods	the effect of switching at t among units with the same treatment history over the last k periods
de Chaisemartin & D'Haultfœuille (2024)	no restriction on carryover; no anticipation; parallel trends of the **status-quo** outcome (period-1 treatment kept)	event-study effects of the *actual* path, ℓ periods after the first change of treatment, against the status-quo path

- Each restriction defines a different parameter: two rows, two different things estimated.
- The two-way fixed-effects coefficient is a weighted sum of $\text{ATT}(\mathbf{d}, t)$ with weights that can be negative (de Chaisemartin and D'Haultfœuille 2020): to the already-treated comparisons of chapter 5 it adds comparisons with units just switched off.

The switchers event study

de Chaisemartin and D'Haultfœuille (2024). Let F_i be the first period where unit i's treatment changes from its period-1 value. For horizon $\ell \geq 1$:

- **Switchers**: units observed ℓ periods after their first switch, $t = F_i - 1 + \ell$, whatever their treatment did in between.
- **Comparison**: units with the same period-1 treatment whose treatment has **not yet changed** at t — the not-yet-switchers of chapter 5, at any baseline level.
- DID_ℓ: the change in outcomes of the switchers from $F_i - 1$ to $F_i - 1 + \ell$, minus the same change for the comparison units, aggregated across switching dates with weights proportional to the number of switchers.

The estimand

The average effect, ℓ periods after the first switch, of having followed one's **actual** treatment path rather than the **status-quo** path (period-1 treatment kept throughout). The path is what is evaluated; current and lagged treatments enter together.

Placebos, normalization, and how to read the curve

- **Placebos** DID_ℓ^{pl}: the same comparison over ℓ periods **before** the switch, with the same units and weights as the effects.
- **Normalized** version: DID_ℓ divided by the cumulative treatment difference between switchers and comparisons, the sum from F_i to t of the gap between their treatments — an effect per unit of treatment, for non-binary treatments or treatments that turn off within the window.
- The horizon ℓ counts periods **since the first switch**, whatever happened to the treatment afterwards. If the treatment is on at $\ell = 1, 2, 3$ and off at $\ell = 4$, DID_4 measures the effect of the path "on for three periods, off for one" — the reversal is inside the estimand.
- A treatment that returns to its baseline within the window makes the non-normalized effect hard to read; the normalized effect is an average per period of exposure and compares across horizons.

The switchers event study: who is compared at each horizon

The curve of the previous slide is not drawn on a fixed set of units. At each horizon ℓ, the switchers and their comparison units are those still available, so the groups change along the curve.

- Comparison units are dropped from the horizon once their own treatment changes: the group changes with ℓ; report its size at each horizon.
- The **same switchers** option keeps the set of switchers fixed across horizons, so the curve is not driven by composition — the balanced-panel check of chapter 4, horizon by horizon.
- Units already treated in period 1 with a different baseline level are compared only with units at the same level: two baselines, two separate DiDs, one aggregate.

Section 3

Workflow

Conclusion: forward engineering, twice

	Potential outcome	Assumption	Estimator
Dose	$Y_{it}(d)$; groups = doses	parallel trends per dose (level curve); strong parallel trends (responses)	binned $\mathrm{ATT}(d \mid d)$ curve and aggregate; ACR only under strong PT
On/off	$Y_{it}(\mathbf{d})$, $\mathbf{d}$ a path; groups = paths	parallel trends of the status-quo outcome; no anticipation; carryover restriction if any	switchers vs not-yet-switchers: DID_ℓ, placebos, normalized

- The diagnostics of chapters 4 and 5 apply cell by cell: pre-trends of each dose bin, each path; a sensitivity bound on the block that carries the result.
- In both cases the two-way fixed-effects coefficient is a weighted average of building blocks, with weights set by the variance of the regressor and comparison cells that may be treated.

Section 4

In R

The switchers event study in R

```r
library(DIDmultiplegtDYN)
# horizons 1..4 after the first switch, 3 placebos;
# comparison = not-yet-switchers, same baseline treatment
est <- did_multiplegt_dyn(
  df, outcome = "y", group = "id", time = "t",
  treatment = "d", effects = 4, placebo = 3,
  cluster = "id", normalized = TRUE,
  same_switchers = TRUE)
summary(est)   # DID_l, placebos, switchers per horizon
```

- `normalized = TRUE` divides each effect by the cumulative treatment difference (the per-period effect); `same_switchers = TRUE` fixes the composition across horizons.
- `controls`, `trends_nonparam` (group-specific trends) and `by` (heterogeneity) exist; each adds an assumption to state.
- The output reports the number of switchers and comparison units at each horizon: put it in the table.

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